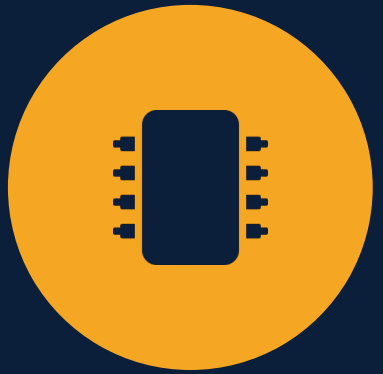
UNIT 3 · 9 hours

# Semiconductor Diode and the PN Junction



Diodes, Transistors and their Applications

**Basic Electrical and Electronics Engineering (EEE103)**
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NMAM Institute of Technology, Nitte · 2025-26

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# Learning objectives

- Describe intrinsic and extrinsic semiconductors
- Explain formation of the depletion region and barrier potential
- Explain forward and reverse bias
- State the diode symbol and its terminals

**Syllabus context**

Semiconductor diode, diode equivalent circuits, half-wave rectifier, full-wave bridge rectifier. Zener diode and its use in voltage regulation.

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# Semiconductors

### Intrinsic

Pure silicon or germanium. Equal numbers of electrons and holes; low conductivity at room temperature.

### N-type

Doped with pentavalent atoms (P, As). Electrons are majority carriers.

### P-type

Doped with trivalent atoms (B, Ga). Holes are majority carriers.

**Conductivity lies between conductors and insulators and rises with temperature and doping.**

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# Formation of the PN junction

1

**Join P and N**

Majority carriers diffuse across the junction.

2

**Recombination**

Electrons and holes recombine near the junction.

3

**Depletion region**

Fixed ions left behind: a region with no free carriers.

4

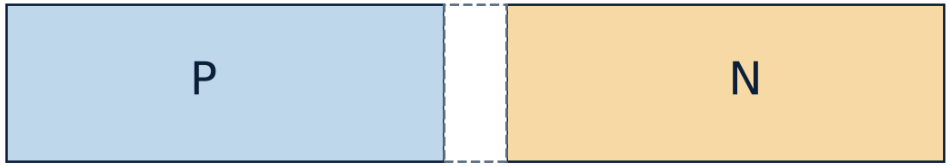
**Barrier potential**

Electric field of the ions stops further diffusion: 0.7 V (Si), 0.3 V (Ge).

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# Forward and reverse bias

Forward bias: depletion region narrows




Reverse bias: depletion region widens




- Forward: P to +, N to −. Applied voltage opposes the barrier; depletion region narrows; current flows when V > 0.7 V.
- Reverse: P to −, N to +. Barrier increases; depletion widens; only tiny leakage (minority carriers).

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# Forward bias versus reverse bias

## Forward bias

- Anode (P) positive
- Depletion region narrow
- Low resistance, large current
- Diode acts as closed switch

## Reverse bias

- Anode negative
- Depletion region wide
- High resistance, μA leakage
- Diode acts as open switch

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# The diode

**Symbol:** triangle (anode) with bar (cathode); current flows from anode to cathode.

**Function:** conducts in one direction only; converts AC to DC.

**Types:** rectifier, signal, Zener, LED, Schottky.

**Applications:** rectification, switching, clamping, protection, voltage regulation.

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# Key takeaways

- N-type: electrons majority; P-type: holes majority
- Depletion region and barrier potential form at the junction
- Forward bias reduces the barrier; reverse bias increases it
- Si turns on near 0.7 V, Ge near 0.3 V

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# Practice questions

**Q1.** Explain how the depletion region is formed.

**Q2.** What happens to the depletion width in forward and reverse bias?

**Q3.** Compare N-type and P-type semiconductors.

**Q4.** Why does a reverse-biased diode still carry a small current?